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Automating social assistance: Exploring the use of robotic process automation in the Swedish personal social services

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Abstract

Many European countries employ Robotic Process Automation (RPA) in the administration of public benefits. However, there is limited understanding of how RPA is applied at the client level. This article investigates the utilization and impact of RPA use on social assistance (SA) distribution in Sweden, drawing on a sample of 800 SA applications in four Swedish municipalities. The results show that RPA use correlates with applicants' country of birth, age and duration of SA receipt. Additionally, RPA implementation coincides with less generous decisions, disproportionately affecting financially vulnerable groups. Rather than a correlation between generosity and the technology itself, the results suggest a conflict between the reorganisation of SA administration during RPA implementation and the principle of individualized judgments inherent in SA casework. Hence, public organisations are encouraged to ensure that their adoption of RPA neither exacerbates unequal access to services nor compromises professional discretion in favour of efficiency-driven measures.

KEYWORDS

automated decision-support, digital automation, robotic process automation, RPA, social assistance, social work

INTRODUCTION

In recent years, European national governments have promoted the adoption of digital automation by public organisations to improve service delivery. While sometimes motivated to enhance transparency and accountability, and provide a more equitable provision of services, the adoption of automated services is primarily driven by the goal of improving internal and external efficiency (Germundsson, 2022; Peláez et al., 2018). The concept of digital automation encompasses a diverse range of services and tools, spanning from highly sophisticated systems that leverage big data for predictive or machine learning purposes (i.e., 'Artificial Intelligence'/AI), to comparatively simpler systems that automate specific administrative processes. While more advanced AI systems are often considered to harbour the highest potential to transform service provision, their practical implementation and use in the context of public administration have also raised significant contention. For example, concerns have been raised over the risk of algorithmic biases (Eubanks, 2018; Gillingham, 2019a), ethical considerations related to privacy and accountability (Gillingham, 2019b; Saldanha et al., 2022), as well as the need for appropriate governance frameworks to effectively address such issues (Miailhe, 2017). Moreover, public service organisations in most European countries have used a variety of digital electronic information systems (EIS) to manage their documentation for several decades (Ylönen, 2022), resulting in a vast amount of unstructured data dispersed over several systems, posing significant challenges in AI-training.

The interplay between different technical functionalities across diverse digital EIS and the complex web of legal and administrative organisational structures has also placed an increased administrative burden on public servants to bridge the administrative technological gaps in and/or between the systems in use. Given both the encouragement to increase the adoption of digital automation, and the need to assist public servants in navigating the intricate terrain of diverse and incompatible IT systems, Robotic Process Automation (RPA) has thus surfaced as an increasingly popular tool within the context of public service. In short, RPA is a type of digital software that operates at the user interface level, where it performs procedural tasks that are usually conducted by humans, either within a single digital system or across multiple systems (Andersson et al., 2022; Lindgren et al., 2019). Although certain RPA systems can incorporate more advanced AI techniques, the most common mode of operation relies on predefined rules and scripts (simple RPA; Gutermuth et al., 2021).

While initially implemented within state-wide organisations responsible for managing and administering universally oriented services and benefits, many countries have also extended their utilization of RPA to encompass local public organisations. One example of this development is the use of RPA in the Swedish Personal Services' handling of Social Assistance (SA). In the Swedish context, social assistance (SA) is a cash benefit aimed at society's most financially vulnerable individuals. Clients' eligibility is assessed on a monthly basis, by caseworkers enjoying a substantial scope of discretion in their decision-making (Stranz, 2019; Tummers & Bekkers, 2014). In this context, *simple* RPA systems are predominantly utilized to facilitate tasks such as data transfer between internal systems, retrieval of information from external organisational registers and the execution of basic calculations. In some municipalities, RPA is also applied to instrumentally estimate the specific amount that a client would be eligible to receive. Despite endorsement and support from central national policy actors,[1] the utilization of RPA in the distribution of SA has sparked several controversies and conflicts, in which RPA has been argued to both jeopardize transparency in the SA eligibility decision-making process, and erode the discretionary room for manoeuvre for caseworkers (Germundsson, 2022; Ranerup & Henriksen, 2020; The Union for Professionals, 2021).

As of yet, research on the practical aspects of implementation and use of RPA in the context of decision-making in public administration is sparse. For example, important aspects yet to be covered include to what extent the technology is actually utilised in SA provision, and whether its use may exert an influence over decisions on SA eligibility. The present article aims not only to address these issues, but also to examine the possible correlation between decisions and factors associated with client background characteristics, labour market attachment and variations in financial circumstances. The article takes an overall explorative approach, utilising data from 800 decisions on social assistance collected in four Swedish municipalities during 2022.

The article addresses the following research questions:

- To what extent do caseworkers utilise RPA support in SA eligibility decision-making?
- Does the use of RPA covary with applicant-specific background factors (e.g., gender, age and household composition), employment status, and/or factors that bear reference to the level of self-support?
- To what extent do the outcomes of SA decisions vary between measurements conducted before and after RPA implementation? If such differences are shown, do they covary with applicant-specific factors?

SOCIAL ASSISTANCE IN THE SWEDISH CONTEXT

A short-term multipart subsidy

In Sweden, the municipally organised Personal Social Services (PSS) have the ultimate responsibility to provide municipal inhabitants with a 'reasonable standard of living' (cf. SFS, 2001:453, the Social Services Act). The benefit is designed as a short-term benefit in the event of municipal inhabitants' temporary inability to financially support themselves, requiring clients to contribute actively to their own self-sufficiency in order to be eligible for support.

The subsidy consists of two main parts, where the *national benefit standard* (NBS) makes up the main portion. The NBS is a sum established yearly by the Swedish national government, defining standardised costs for a specific set of household expenses such as food, clothing and toiletries. Secondarily, PSS organisations specify their

[1] E.g., the Swedish Association of Local Authorities and Regions (SALAR) and the National Board of Health and Welfare (NBHW).

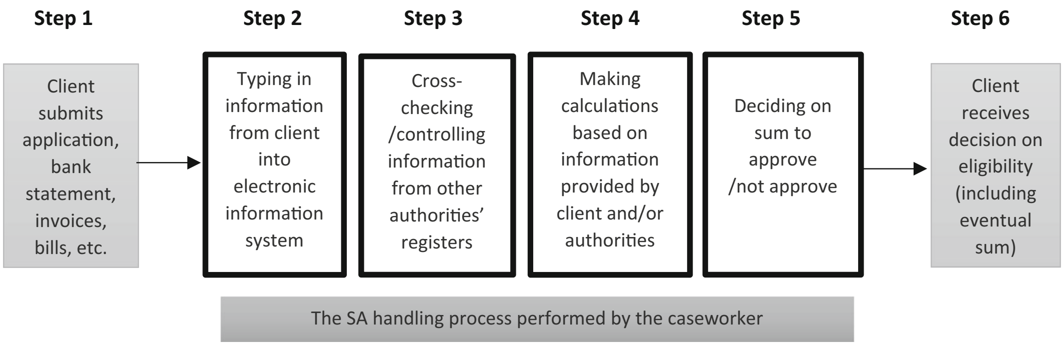

FIGURE 1 Traditional monthly handling process of SA applications. (In this illustration, the first and last steps (1 and 6) are intended to reflect the aspects that involve the client (submitting their application and receiving their decision), while Step 2–5 illustrate the actions taken by the caseworker when administering the application.)

accepted costs for rent, heating and other bills in a set of *municipal standards* that are based on estimated local averages as well as the policies of the municipality in question (Kjellbom & Lundberg, 2018). In addition to these two categories of continuing costs, support for more sporadic expenses—often referred to as *additional assistance*—can be applied for. While clients have the legal right to apply for anything they need, additional assistance most often covers the costs of dental care, eyeglasses and identification cards.

SA casework as a twofold mission

SA caseworkers are academically trained professionals, mainly responsible for motivating and supporting clients in their efforts to become self-sufficient (Minas et al., 2018). Such work can be practically oriented: for example, directing clients towards unemployment services or aiding them in obtaining a medical certificate for sick leave, advising them to apply for other appropriate governmental benefits, and/or guiding them towards municipal departments for financial advice. Perhaps even more importantly, caseworkers are also tasked with drawing attention to clients' possible needs for more socially oriented support, including facilitating contacts with other units within the PSS, attending meetings with healthcare providers, and/or generally motivating clients' efforts toward self-sufficiency.

In parallel with these efforts, SA caseworkers are generally responsible for a set of tasks related to investigating eligibility for the cash benefit and organising its distribution in practice. This process, illustrated in Figure 1, recurs each month upon the client's submission of their completed application, wherein they specify their personal information (e.g., living conditions, marital status, children) as well as their expenditures and income. In addition, claimants are generally obliged to submit documentation supporting their SA claims: bank account statements showing their lack of funds and the bills they seek assistance in paying, along with information about their efforts to become self-sufficient; either a medical certificate of their inability to work or a list of a predefined number of applied jobs; and/or attendance at measures to which they have been assigned (Step 1). The caseworker enters the information provided by the client into the EIS (Step 2) before cross-checking and/or gathering additional information from other relevant authorities (Step 3) and making calculations based on the information at hand (Step 4). The process then concludes with caseworkers deciding on the clients' eligibility for support (Step 5), before finally informing them of the decision outcome, including the size of any sum granted (Step 6).

RPA IN THE SA CONTEXT

Based on municipalities' discretion to organise and conduct their work in line with their specific political and organisational circumstances, local authorities have made varying choices regarding the incorporation of RPA in different stages of the social assistance (SA) handling process. Moreover, the design of the software differs between municipalities, based both on their currently procured EIS as well as the additional providers procured to deliver the RPA. This means that while some municipalities' RPA simply transfers client data from electronically submitted applications into the EIS (i.e., Step 2), others have programmed an RPA that can take on one or several consecutive steps in processing the application. Although several municipalities allow the RPA to propose a decision to caseworkers (i.e., Step 4), all Swedish municipalities who utilise RPA except one[2] have opted to retain the caseworker as a 'human-in-the-loop' in the

[2]This municipality is not included in this study.

process. This involves reviewing and approving, or disapproving, the suggested decision before communicating it to the client.

Furthermore, the practical possibility to utilise simple RPA technologies in the context of SA is conditioned by three main criteria. First, the client's application needs to be *digitised*, meaning that it must be either submitted digitally, or manually converted by a caseworker into digital data. Moreover, the form of RPA currently utilised in Swedish municipalities is unable to handle any type of information provided outside of the EIS in which it is programmed to work. As such, clients submitting a digital SA application have been excepted from the need to regularly provide bills and bank statements to prove their lack of funds. Second, the data must be *standardised*—typed and formatted in accordance with the specific configuration recognized by the RPA system implemented in the respective municipality. As a result, incorrectly filled out applications, occurrences of misspelled words and similar errors necessitate manual processing. Third, the content of the application needs to be fairly *uncomplicated* in order to fit the exact rule-based process that the RPA software is programmed to execute. This means that cases where specific individual considerations or exceptions are needed must be processed separately. Subsequently, for SA applications where one or more of these criteria are not met, a caseworker must either correct or fully process the application manually.

Another important prerequisite for utilising RPA in SA casework is the argument that it can be divided into administrative tasks that *can* be automated, and supportive aspects that *cannot*. Therefore, the practical implementation of RPA in the context of SA involves determining the actions that are considered sufficiently rule-based to be presented in a sequential manner and thus deemed amenable to algorithmic representation. In addition, several municipalities have opted to combine the RPA implementation with a reorganisation of caseworkers, assigning them to either administratively oriented units or units focusing on supportive tasks, with a varying degree of organisational detachment between the two.

RPA in SA casework from a socio-technical lens

The use of various forms of digital tools in the social work context has sparked growing international interest in how they might impact both the efficiency of work processes and the effectiveness of outcomes (Considine et al., 2022; Jørgensen et al., 2022). For example, a number of studies have focused on different forms of automated systems intended to assist in more objective or 'unbiased' decision-making and how professional discretion might be reconfigured by their use (cf. Busch et al., 2018; Coulthard et al., 2020; Høybye-Mortensen, 2015; Meilvang & Dahler, 2022; Zouridis et al., 2020). The results have pointed in different, sometimes contradictory, directions. Some have claimed that caseworkers' professional discretion is safeguarded by the complexity of clients' situations (Petersen et al., 2020), while others have claimed that the codification process involved in creating automated systems contributes to discounting the complexity of casework (Andersson et al., 2022). Case studies focusing on the use of RPA in the Swedish SA context have also produced somewhat conflicting results, suggesting that caseworkers remain in close contact with clients while simultaneously increasing the efficiency of their SA administration (Svensson, 2019), but also that RPA has weakened the caseworker's primary role as a decision maker (Ranerup & Henriksen, 2020). Moreover, it has been argued that RPA causes stricter rule-following in the decision-making process (Ranerup & Svensson, 2023), but also that decision outcomes do not necessarily become more ethical or equal (Devlieghere et al., 2017).

In this study, SA caseworkers are considered street-level bureaucrats, who are granted significant discretion when applying national legislation, municipal policies and local organisational practices in handling SA applications (cf. Lipsky, 2010). Taken together with the twofold mission of providing clients with both social support and motivation while simultaneously investigating and administering the cash benefit, caseworkers also exercise a high degree of discretion in considering the client's particular circumstances and their efforts to become self-sufficient when determining SA eligibility. While steps 2 through 4 (as illustrated in Figure 1) are primarily framed as administratively oriented, caseworkers are hence also tasked with evaluating the client's active efforts to progress towards self-sufficiency throughout the decision-making process (Bernhard & Wihlborg, 2022; De Witte et al., 2016). The combination of caseworkers' multifaceted activities situates them in a complex—sometimes even contradictory—position of navigating discretional practical judgements and legal and moral issues, as well as mastering challenging interactions with clients and other administrators (cf. Wagenaar, 2004). This means that discretional space is not only present in the caseworkers' final determination on a client's eligibility but is intricately interwoven throughout the administratively oriented process and in their considerations of the specific client's individually adapted efforts to achieve self-sufficiency. Based on this, the distinct separation of professional SA casework between motivational and supportive aspects on one hand, and the administrative aspects on the other, should primarily be viewed as a theoretical construct.

While the implementation of RPA is typically accompanied by a reorganisation of organisational and practical aspects in SA casework, digital technologies are not here regarded as neutral tools with predetermined effects within implementing organisations. Instead, they are seen as dynamic artefacts that are shaped both by their own materiality and how individuals and organisations respond to and interact with them (Leonardi, 2012). From this point of view, technologies are continuously constructed, not only by their creators and their socially attributed value and purpose, but also by their inherent possibility to be updated, modified and scaled (Plesner et al., 2018; Tilson et al., 2010; Yeung & Lodge, 2019). As such, the material forms of RPA utilised in Swedish PSS organisations is regarded as recursively entangled with the particular culture and logic of the contexts in which they are programmed, as well as the specific settings where they are implemented (Parton, 2008).

METHODS

The results in the present paper were based on cross-sectional data that aimed to reflect the influence of RPA support in assessments of SA eligibility. The sample comprised information from a total of 800 households that applied for SA, one half during a given month (October 2017, 2018 and 2019 respectively) the year *before* RPA support was implemented, and the other half during the corresponding month (October 2021)[3] *after* RPA support had been adopted. Data were collected in four Swedish medium-sized municipalities (ranging between 29,000 and 95,000 inhabitants) in 2022. The dataset was equally distributed among the municipalities, consisting of 100 cases each for both pre- and post-implementation of RPA per municipality. Ethical approval for the project was obtained from the Swedish Ethical Review Authority (ref. 2021-04017; 2022-03469-02).

The municipalities were pragmatically selected among those that had implemented and made use of RPA support in SA eligibility assessments at least 6 months prior to *post* measurement, that is, April 2021. The six-month limit was based on the assumption that organisations need a certain amount of time to fully implement new work procedures. Thus, to prepare for data collection, a survey of RPA support in Swedish municipalities ($N = 290$) was conducted in spring 2021. The survey showed that 24 municipalities had implemented some sort of RPA support at the time of measurement. Among these, 16 municipalities did not match the set criteria for inclusion (i.e., RPA had been in use for too short a period, the municipality had less than 25,000 inhabitants or SA administration was divided between city districts[4]). PSS managers (or their equivalents) in the remaining eight municipalities were contacted and provided further information about the aim and overall set-up of the project.

A total of four municipalities verbally agreed to participate in the project, after which we formally applied to the local welfare boards to collect data directly from registers of SA applicant households. In each respective municipality, two samples were drawn from local registers using the random function in Excel. The samples comprised applicants of working age (18–64 years old), one covering the *post-group* and one covering the *pre-group*. As shown in Table 1, both subsamples represented about 14% of all decisions on SA during pre- and post-measurements respectively.

Data were collected via on-site visits, where two members from the research team spent five working days in each municipality, reading and transferring data from electronic as well as analogue case files to SPSS files. Although no data that enabled identification of individuals/households were gathered, all files were kept encrypted. Data covered a broad variety of variables on the household level. In addition to information about applicant backgrounds (e.g., gender, age and ethnicity), data were also collected about household composition, educational level and employment status, for example. Further, SPSS sheets also included detailed information about income and expenditures as well as specifics on certain items included in SA applications during the given month(s).

As already pointed out, the design of systems for RPA support may vary between organisations. A common feature for all municipalities covered in the present study was that the RPA transferred data from clients' electronic applications into the operating systems and that it presented some sort of calculation indicating whether the applicant met the criteria for the National Benefit Standard (NBS) or not.

Analyses

Data were analysed on bivariate and multivariate levels using SPSS version 29. The analyses covered (a) *RPA usage* and (b) *decision outcomes*, respectively. Whereas analyses of (a) only covered post-measurement, separate

[3]As the year for implementation of RPA-support varied between municipalities, the point of measurement with respect to the first half of the dataset did as well (cf. Table 1).

[4]In order to ensure a statistically relevant household sample, smaller municipalities were excluded. If SA administration is divided among different organisations, implementation processes and the way in which the RPA is used may differ.

TABLE 1 The sample, municipality size, year of RPA implementation and sizes of SA populations.

Municipality	Inhabitants	RPA, year implemented	Pre-group, *N*	Post-group, *N*
A	95,000	2018	998	1095
B	65,000	2018	704	641
C	29,000	2020	519	422
D	52,000	2019	558	662

Note: Data on population size have been obtained from Statistics Sweden (2021).

TABLE 2 Independent variables utilized in the multiple regression analyses.

	Pre-group[a]	Post-group[a]	Total[a]
Gender, male (%)	54.3	55.4	54.9
Foreign born (%)	64.4	63.6	64.0
Age, mean (SD)	42.4 (13.0)	42.6 (13.3)	42.5 (13.1)
<34 years old (%)	30.1	31.7	30.9
35–49 years old (%)	33.7	33.2	33.5
50+ years old (%)	36.2	35.1	35.6
Single (%)	79.9	79.1	79.6
No children in the household (%)	57.7	65.4	61.8
Months since 1st application, mean (SD)	43.4 (50.9)	60.7 (60.7)	52.6 (56.9)
1–6 months (%)	17.5	9.2	13.1
7–12 months (%)	14.4	7.9	10.9
13–59 months (1–5 years) (%)	45.1	43.5	44.3
60–months (5+ years) (%)	23.0	39.3	31.8
Unemployed (%)	70.2	74.4	72.5
Did not apply for SA previous month (%)	14.4	12.9	13.6
No additional income in previous month (%)	36.5	43.8	40.4
No additional income in current month (%)	35.3	41.2	38.4
Application within the NBS (%)	82.5	81.5	82

Note: Numbers represent percentages and means, with standard deviations in parenthesis (SD).
[a]$n = 326$ (pre-group), $n = 379$ (post-group) and $n = 705$ (total).

analyses of (b) were carried out for both pre- and post-groups.

After clearing data for non-response items, RPA usage (a) was analysed by means of multivariate logistic regression analysis ($n = 378$), focusing on the significance of factors on the individual level (cf. Table 3 for application of RPA support (no = 0 [ref.], yes = 1). Decision outcomes (b) (full approval/partial denial/denial) were first analysed on the bivariate level ($n = 705$) and statistically tested using Pearson's χ^2-test. These outcomes were then examined using multiple logistic regression analyses, focusing on the influence that the same factors as in the analysis of (a) (cf. Table 5) exerted on decision outcomes *before* ($n = 291$) and *after* ($n = 337$) RPA was adopted. The multivariate analyses were cleared for observations that were fully denied ($n = 77$) and, hence, focused on partial approvals (full approval = 0 [ref.], partial denial = 1). In order to control for unobserved heterogeneity on the municipal level, municipality dummy variables were included in all multivariate logistic regression analyses. The distribution of the independent variables is presented in Table 2.

As shown in Table 2, both the sub-samples and the overall sample were dominated by applicants born outside of Sweden, single households and households without children. About three quarters of the applicants were registered as unemployed. All in all, these figures correspond well with previous mappings of SA applicants (cf. Stranz et al., 2016). Further, the distribution of independent variables presented a sample which, in all respects, seemed quite dependent on SA benefits. For example, a significant portion of households, about 75%, had applied for SA for the first time over a year ago, and around 40% of applicants lacked additional income during the month of measurement and/or the month prior.

Table 5 shows an overall similarity in the characteristics of the subsamples. However, noteworthy differences were observed regarding *months since first application*, where the data for the post-group indicated an even more entrenched dependency on SA, as well as presence of *additional incomes*. The latter were less prevalent within the post-group. Moreover, within the post-group, there was a slightly higher share of households without children.

RESULTS

When RPA had been implemented for at least 1 year, it was utilised to process 60% of SA applications. As shown in Table 3, applicants' country of birth, age and time since first applying for SA were all associated with the use of RPA. While applicants born in Sweden were far more likely to be administered via RPA, the use of RPA decreased substantially with client age. Odds ratios (ORs) were also higher for those whose first application was further back in time.

Since RPA use is conditioned on a *digitised* application, these results are likely associated with younger peoples' higher digital competence, as well as their greater material access to computers, smartphones and electronic identification (cf. Bernhard & Wihlborg, 2022; Breit & Salomon, 2015; Hansen et al., 2018; van Deursen & van Dijk, 2019). Moreover, the fact that digital application services are exclusively offered in Swedish, in combination with the fact that RPA is only able to process *standardised* applications (i.e., without spelling errors or incorrect numbers), could also help explain why foreign-born clients had significantly lower odds of automated case processing. Concerning the more longstanding cases' higher odds of RPA utilisation, these tendencies seem less likely to be associated with clients' methods of submitting their application, and more likely to be related to aspects that are more common to prolonged SA recipiency. For example, access to the more universally oriented Swedish welfare programs is closely linked to previous or current employment on the regular labour market. In general, clients receiving SA for longer periods are often more persistently unable to gain employment, and thus less likely to escape SA as their main source of income (cf. Bergnéhr, 2016; Marttila et al., 2010). In such cases, circumstances are often relatively consistent over time, which could render them particularly amenable to automation.

Table 4 shows the overall proportions of decision outcomes pre- and post-RPA implementation. While the share of fully denied applications remained about the same, there was a significant decrease in applications that were fully approved. Before implementation of RPA, just under 45% of applications were defined as partially approved; after implementation, that number increased to almost 60%. Hence, pre- and post-measurements indicate that implementation of RPA coincided with an overall increased restrictiveness.

Assessment outcomes are further analysed in Table 5, which presents ORs for client-specific factors that covary

TABLE 3 Use of RPA.

	Use of RPA[a]	
	OR	**CI (95%)**
Gender (ref. male)	1.19	0.71–1.97
Country of birth (ref. foreign-born)	2.16**	1.25–3.74
35–50 y/o (ref. <34 years old)	0.46*	0.24–0.86
50+ y/o (ref. <34 years old)	0.41**	0.21–0.80
Marital status (ref. single)	1.19	0.56–2.51
Children in household (ref. no)	0.92	0.49–1.75
Employment status (ref. unemployed)	0.99	0.58–1.72
1st application 7–12 months ago (ref. <6 months ago)	3.17*	1.01–9.95
1st application 13–59 months ago (ref. <6 months ago)	2.90*	1.19–7.05
1st application 60+ months ago (ref. <6 months ago)	3.29*	1.28–8.42
SA previous month (ref. no)	1.68	0.81–3.51
Additional income previous month (ref. no)	1.70	0.83–3.77
Additional income current month (ref. no)	0.64	0.28–1.42
Application for additional SA (ref. no)	1.04	0.57–1.19

Note: Numbers reflect odds ratios (OR) and confidence intervals (CI).
[a] $n = 378$.
p*: † < 0.10; * < 0.05; ** < 0.01; * < 0.001.

TABLE 4 Proportions of cases receiving full approval, partial denial or full denial, before and after implementation of RPA.

	Before RPA	After RPA	*p*[b]
Approved[b]	45.1	31.7	***
Partially denied[b]	44.2	57.5	***
Denied[b]	10.7	10.8	ns

Note: Numbers reflect percentages.
[a]Numbers are statistically tested using Pearson's χ^2-test.
[b]$n = 267$ (full approval), $n = 362$ (partial denial), $n = 76$ (full denial).
* † < 0.10; * < 0.05; ** < 0.01; *** < 0.001.

TABLE 5 Partial denial, before versus after RPA implementation.

	Before RPA[a]		After RPA[a]	
	OR	CI (95%)	OR	CI (95%)
Gender (ref. male)	1.25	0.69–2.26	1.15	0.66–1.99
Country of birth (ref. foreign-born)	1.03	0.57–1.85	0.59†	0.33–1.03
35–50 years old (ref. <34 years old)	1.92†	0.96–3.85	1.72	0.88–3.34
50+ years old (ref. <34 years old)	0.83	0.41–1.70	2.77**	1.34–5.69
Marital status (ref. single)	3.55**	1.52–8.28	4.38*	1.43–13.42
Children in household (ref. no)	0.64	0.33–1.23	1.26	0.61–2.61
Employment status (ref. unemployed)	1.03	0.56–1.91	0.83	0.46–1.53
1st application 7–12 months ago (ref. <6 months ago)	0.29*	0.11–0.80	1.45	0.41–5.14
1st application 13–59 months ago (ref. <6 months ago)	0.48†	0.21–1.08	0.88	0.34–2.28
1st application 60+ months ago (ref. <6 months ago)	0.52	0.21–1.32	1.04	0.39–2.93
SA previous month (ref. no)	1.78	0.67–4.69	1.09	0.37–3.22
Additional income previous month (ref. no)	1.03	0.42–2.55	0.64	0.28–1.46
Additional income current month (ref. no)	3.43*	1.33–8.87	1.64	0.70–3.83
Application for additional SA (ref. no)	1.87†	0.92–3.80	3.93*	1.72–8.98
Use of RPA (ref. RPA not used)	n/a	n/a	1.02	0.59–1.78

Note: Multiple logistic regression analyses (ref. full approval). Numbers reflect odds ratios (OR) and confidence intervals (CI).
[a]$n = 291$ (before RPA) and $n = 337$ (after RPA).
* † < 0.10; * < 0.05; ** < 0.01; *** < 0.001.

with partial denials before and after the implementation of RPA. As shown, before RPA, cohabiting/married applicants have significantly higher odds of receiving a partial denial rather than a full approval, which is congruous with the maintenance obligation between members of cohabiting and/or married couples that is applied by Swedish PSS organisations (Hussénius, 2023). The same tendency—possibly associated to those cohabiting—can be seen for households with some form of income besides SA. This is also consistent with the 'last resort' nature of the SA benefit, which requires the caseworker to consider all other (possible) sources of income when investigating the household's right to assistance. It is conceivable that applicants with smaller additional incomes may inadvertently overlook them when applying for SA. As a result, they may unintentionally apply for amounts that exceed the NBS. Finally, there is a pattern associated with the amount of time since the client's first SA application. In comparison to those who made their first application up to 6 months before the pre-measurement month, those who first applied for SA 7–12 months prior had higher odds of receiving a full approval of their application. The same tendency can be seen even in relation to more longstanding cases; however, these results are not significant.

After the implementation of RPA, the most prominent result is that clients applying for *additional assistance* had significantly higher odds of a partially denied (rather than a fully approved) application. This means that clients facing particularly vulnerable circumstances—whose expenses exceed the standardised amounts provided by the NBS and the municipal standards—have higher odds of not receiving the full sum they applied for. Based on the significance displayed, it is also likely that the partial denial pertains specifically to the additional assistance portion of the application.

Albeit weakened, the odds of having their application partially denied increased for cohabiting/married applicants after RPA implementation. While the previously higher odds of partial denial for households with additional incomes did not persist, they increased for clients in the highest age groups (50+ years). It should also be noted that before RPA, there were no differences in the odds between native-born and foreign-born clients. After RPA, however, native-born clients had lower odds of receiving a partial denial of their application, compared to foreign-born clients. While fairly close, this tendency does not reach significance and should thus be interpreted with caution.

Notably, no significant differences were found relating to the variable reflecting the use of RPA. Therefore, we cannot attribute neither the increased restrictiveness in Table 3 nor the various effects in Table 4 solely to the use of RPA. Based on the design of this study, shifting application patterns or other aspects relating to client behaviours cannot be ruled out. However, the shifts in approval rates seem most likely to be related to the comprehensive efforts to reorganise and restructure SA administration in conjunction with the implementation of RPA.

DISCUSSION

This article has aimed to explore and analyse the extent to which Swedish SA caseworkers utilise RPA to support their decision-making, the extent to which RPA use relates to decision outcomes, and whether any of these questions covary with applicant-specific factors. The results show that when RPA had been implemented for at least 1 year, it was used in a majority of SA cases; however, the odds of RPA use were significantly higher for applicants born in Sweden, applicants up to 34 years old and longstanding applicants. Based on previous research, these outcomes presumably correspond with the ability of certain subgroups to submit a digital SA application that conforms with the streamlined procedure for which the RPA is programmed.

The divide among demographic groups regarding access and utilization of digital services, has been identified as a significant concern within social work contexts, as it can exacerbate social exclusion for groups already facing disadvantages (Eubanks, 2011; Steyaert & Gould, 2009). Pertaining to RPA use, this divide seems to primarily manifest between the 'typical' cases which government programs mainly target and the 'atypical' cases, placing the latter at heightened risk of exclusion from these programs (Larsson, 2021). Similarly, digital application processes have been identified as impersonal and inadequate in response to complexities that clients might face (Hetling et al., 2014).

Furthermore, consistent with the requirement for a *digitised* application, all four municipalities in this study incorporated the implementation of RPA with the discontinuation of requesting bank statements and bills as evidence of insufficient funds. Importantly, however, this exemption did not extend to clients submitting paper applications, which could be seen as placing a disproportionately higher burden on digitally disadvantaged individuals to provide evidence and disclose their personal circumstances during the SA application process. Such administrative burdens, which generally tend to affect citizens in already disadvantaged groups, may ultimately prevent citizens from accessing benefits or public services to which they might be entitled (Brodkin & Majmundar, 2010; Larsson, 2021; Lindgren et al., 2019; Moynihan et al., 2015; Schou & Pors, 2019). Moreover, RPA requirements for standardised and simplified applications could also contribute to reducing clients' possibilities for communicating their individual circumstances.

As regards approval rates, our results show that implementation of RPA coincides with a pronounced restrictiveness. This is particularly true for married and/or cohabiting clients, clients 50 years or older, and those applying for additional assistance. Although not significant, a tendency to greater restrictiveness towards foreign-born clients can also be seen. Importantly, however, no significant differences in approval rates were found in relation to the utilisation of RPA, which supports the impression that the restrictiveness is associated with organisational and political rearrangements linked to the technology's implementation, rather than the technology itself. Nonetheless, the RPA's requirement for a codifiable context could also be contributing to the configuration of SA casework to align with the simplified, linear and efficiency-oriented qualities associated with the technology (Andersson et al., 2022). For example, despite the discretion to diverge from a digitally produced suggestion, caseworkers have been shown to be disinclined to critically assess recommendations presented by a digital decision-suggestion system (Enarsson et al., 2022). Furthermore, de Boer & Raaphorst, 2023 have argued that while the implementation of automation may not explicitly restrict caseworker discretion, street-level bureaucrats utilizing decision-support systems exhibit diminished discretion compared to their counterparts who do not. This suggests that caseworkers could be directed towards adhering to preprogrammed rules embedded within the automated system, rather than exercising their discretion to interpret the rules in light of individual clients and their specific circumstances (de Boer and Raaphorst, 2023). Based on both previous studies and the findings presented here, the use of automated services could thus be conducive to caseworkers taking a more instrumental stance towards their own decision-making: allowing the technology to 'find out' whether the household is eligible for SA, rather than discretionally interpreting regulations and guidelines in order to produce the 'fact' of client eligibility (Hussénius, 2023). As such, one possible interpretation of our results is that caseworkers might become less inclined to exercise their professional discretion to consider the vulnerability of particular client groups after RPA has been implemented in SA casework.

While lessened consideration of individual client characteristics may aid in reducing unequal outcomes of decision-making, digital frontline work has also been related to caseworkers experiencing 'alienation' vis-à-vis clients and their individual situations (Løberg & Egeland, 2021). The present data does not allow for establishing a direct correlation between automation and alienation in the social work context. However, the results indicate that the comprehensive, discursive and practical adjustments associated with the implementation of RPA in SA management could be associated with a less pronounced focus on the situation-specific care and qualitative judgements traditionally considered integral aspects of social work practice (cf. Andersson et al., 2022; Green, 2022). Combining these insights with the tendency for caseworkers to defer to automated systems – that is, reducing the amount of independent scrutiny that they exhibit when making decisions (Green, 2022)—the common organisational response of safeguarding against diminished discretion in public administration by keeping a 'human-in-the-loop' seems an insufficient measure. Rather, organisations turning to automation to better serve citizens must take an active stance to ensure not only their desired outcomes, but also that clients are not asymmetrically affected.

This study has shown that while RPA in the SA context might act as a tool for internally streamlining administrative processes in organisations, it might simultaneously create unevenly distributed restrictions on citizens' access to the SA benefit. In conclusion, then, the conventional view of RPA as an unbiased administrative system that allows caseworkers to better support client self-sufficiency can also obscure the possibility for its implementation to impact on the outcomes of SA decisions. Rather, the combined consequences of the RPA software's inherent requirements for digitising, standardising, and simplifying SA casework, and the organisational adjustments with which its implementation tend to intertwine seem at odds with the logic of individual and qualitative judgements integral to SA casework. As such, PSS organisations opting to adopt RPA in their administration of SA should not only exercise caution to prevent the perpetuation of existing disparities in access to their services, but be mindful not to inadvertently give up their own expertise and discretion in the pursuit of increased efficiency.

Limitations

The presented data possesses some notable strengths. The data collection procedure merits recognition, as it entailed on-site visits that not only facilitated direct validation of any uncertainties pertaining to specific items with caseworkers, but also facilitated a comprehensive understanding of the operational functionalities of local RPA configurations.

However, the presented data also has limitations that should be highlighted. Firstly, cross-sectional designs, such as the one used in this study, not only provide information from a given moment in time, but also tend to encompass a larger proportion of individuals with substantial and complex problems compared to longitudinal designs. In this specific case, this may have introduced a negative bias in our analysis of the actual use of RPA. It is reasonable to assume that automated systems are better suited for decision-making scenarios that involve more predetermined outcomes. This assumption is partially supported by the observation that RPA usage covaries with the duration of SA recipiency. Secondly, despite the inclusion of municipalities representing 25% of those who have implemented RPA within the sampling frame, both the sub-samples and the overall sample are noticeably small.

Consequently, in line with our exploratory approach throughout the article, our findings should be interpreted with caution. Our analyses do not support definitive conclusions regarding a direct causality between RPA usage and decision outcomes. Instead, the results should be regarded as tentative, underscoring the necessity for future studies designed with an evaluative approach.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ETHICS STATEMENT

Ethical approval was obtained from the Swedish Ethical Review Authority (ref. 2021-04017; 2022-03469-02).

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